



CLOUD COMPUTING

Assignment 01

Gitea • GitHub • Git LFS • GitHub Pages

Department	Software Engineering
Assignment No	01
Subject	Cloud Computing
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Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

2. Update the Ubuntu package index and install Docker Engine and curl. On Ubuntu, the required package name is docker.io:

```
sudo apt update -y
sudo apt install -y docker.io curl
```

```
Last login: Thu Sep 17 16:24:41 2026 from 192.168.124.1
engineer-sidra-iftikhar@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
engineer-sidra-iftikhar@ubuntu:~$ |
```

Fig. 1: Docker version check on the Ubuntu server

3. Create the system-wide Docker CLI plug-in directory and download the Docker Compose plug-in for an x86_64 Ubuntu server:

```
sudo mkdir -p /usr/local/lib/docker/cli-plugins sudo curl -SL
https://github.com/docker/compose/releases/latest/download/docker-compose-linux x86_64 -o
/usr/local/lib/docker/cli-plugins/docker-compose
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
```

```
engineer-sidra-iftikhar@ubuntu:~$ sudo mkdir -p /usr/local/lib/docker/cli-plugins
[sudo: authenticate] Password:
engineer-sidra-iftikhar@ubuntu:~$ sudo curl -SL ... -o /usr/local/lib/docker/cli-plugins/docker-compose
% Total    % Received % Xferd  Average Speed   Time    Time     Time    Current
           Dload  Upload  Total   Spent    Left     Speed
0         0     0      0      0      0      0      0      0      0      0      0      0      0      0      0      0      0      0      0      0
0curl: (6) Could not resolve host: ..
engineer-sidra-iftikhar@ubuntu:~$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
chmod: cannot access '/usr/local/lib/docker/cli-plugins/docker-compose': No such file or directory
engineer-sidra-iftikhar@ubuntu:~$ |
```

Fig. 2: Docker Compose plug-in download attempt (network error)

4. Start the Docker service and add your current Ubuntu account to the docker group:

```
sudo systemctl start docker
sudo usermod -aG docker $USER
exit
```

Completely sign out of the Ubuntu server and reconnect so that the new group membership takes effect. Opening only a new terminal tab is not sufficient.

```
engineer-sidra-iftikhar@ubuntu:~$ sudo systemctl start docker
engineer-sidra-iftikhar@ubuntu:~$ sudo usermod -aG docker $USER
engineer-sidra-iftikhar@ubuntu:~$ exit|
```

Fig. 3: Docker service started and user added to docker group

5. Verify the Git, Docker, and Docker Compose installations:

```
ssh <github-username>@<ubuntu-server-ip>
git --version
docker --version
docker compose version
```

Do not continue until all three commands display their installed versions.

```
engineer-sidra-iftikhar@ubuntu:~$ git --version
git version 2.53.0
engineer-sidra-iftikhar@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
engineer-sidra-iftikhar@ubuntu:~$ docker compose version
docker: unknown command: docker compose

Run 'docker --help' for more information
engineer-sidra-iftikhar@ubuntu:~$ |
```

Fig. 4: Git and Docker versions verified

```
engineer-sidra-iftikhar@ubuntu:~$ ls -lh /usr/local/lib/docker/cli-plugins/
total 0
engineer-sidra-iftikhar@ubuntu:~$ sudo mkdir -p /usr/local/lib/docker/cli-plugins
[sudo: authenticate] Password:
engineer-sidra-iftikhar@ubuntu:~$ sudo curl -SL https://github.com/docker/compose/releases/latest/download
4 -o /usr/local/lib/docker/cli-plugins/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload  Total  Spent  Left  Speed
  0     0    0     0    0     0      0  0     0      0  0     0      0
  0     0    0     0    0     0      0  0     0      0  0     0      0
100 31.14M 100 31.14M    0     0  215.6k      0  02:27  02:27    246.5k
engineer-sidra-iftikhar@ubuntu:~$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
engineer-sidra-iftikhar@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
engineer-sidra-iftikhar@ubuntu:~$ |
```

Fig. 5: Docker Compose plug-in installed successfully (v5.6.0)

6. Clone the instructor-provided Gitea setup repository on the Ubuntu server:

```
cd ~
git clone https://github.com/WaqasSaleem97/Gitea.git
cd Gitea
```

```
engineer-sidra-iftikhar@ubuntu:~$ cd ~
engineer-sidra-iftikhar@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Gitea.git
Cloning into 'Gitea'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 2.69 MiB/s, done.
Resolving deltas: 100% (4/4), done.
engineer-sidra-iftikhar@ubuntu:~$ cd Gitea
engineer-sidra-iftikhar@ubuntu:~/Gitea$ |
```

Fig. 6: Cloning the Gitea setup repository

7. Use the existing compose.yaml without changing its server URL or port settings. Start Gitea and PostgreSQL:

```
docker compose up -d
```

```
engineer-sidra-iftikhar@ubuntu:~$ cd Gitea
engineer-sidra-iftikhar@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
✓Image postgres:17-alpine Pulled
✓Image gitea/gitea:latest Pulled
✓Network webnet Created
✓Volume gitea_postgres Created
✓Volume gitea Created
✓Container gitea Started
✓Container gitea_db Started
engineer-sidra-iftikhar@ubuntu:~/Gitea$ |
```

Fig. 7: Gitea and PostgreSQL containers started

8. Confirm that both containers are running:

```
docker compose ps
```

9. Confirm that Gitea responds locally on port 3000:

```
curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
```

```
engineer-sidra-iftikhar@ubur X + v
engineer-sidra-iftikhar@ubuntu:~/Gitea$ docker compose ps
NAME      IMAGE          COMMAND                  SERVICE    CREATED        STATUS        PORTS
gitea     gitea/gitea:latest "/usr/bin/entrypoint..." gitea      10 minutes ago Up 10 minutes 22/tcp, 0.0.0.0:3000->3000/tcp, [
::]:3000->3000/tcp
gitea_db  postgres:17-alpine "docker-entrypoint.s..." db         10 minutes ago Up 10 minutes 5432/tcp
engineer-sidra-iftikhar@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
engineer-sidra-iftikhar@ubuntu:~/Gitea$ |
```

Fig. 8: Containers running and Gitea HTTP status 200

10. Display the Ubuntu server's IP address:

```
ip addr
```

```
engineer-sidra-iftikhar@ubuntu:~/Gitea$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
   link/ether 00:0c:29:2b:4b:c4 brd ff:ff:ff:ff:ff:ff
   altname enp2s1
   altname enx000c292b4bc4
   inet 192.168.124.132/24 metric 100 brd 192.168.124.255 scope global dynamic ens33
       valid_lft 1304sec preferred_lft 1304sec
   inet6 fe80::20c:29ff:fe2b:4bc4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Fig. 9: Ubuntu server IP address

Step 1.2: Create and Push the Initial Repository

1. In Gitea, create an empty public repository named exactly Assignment01.
2. Do not initialize it with a README, .gitignore, or license.

New Repository

A repository contains all project files, including revision history. Already hosting one elsewhere? [Migrate repository.](#)

Owner  engineer-sidra-iftikhar

Some organizations may not show up in the dropdown due to a maximum repository count limit.

Repository Name

Good repository names use short, memorable and unique keywords. A repository named ".profile" or ".profile-private" could be used to add a README.md for the user/organization profile.

Visibility ☐ Make repository private

Only the owner or the organization members if they have rights, will be able to see it.

Description

Template

Issue Labels

Fig. 10: New Repository form in Gitea

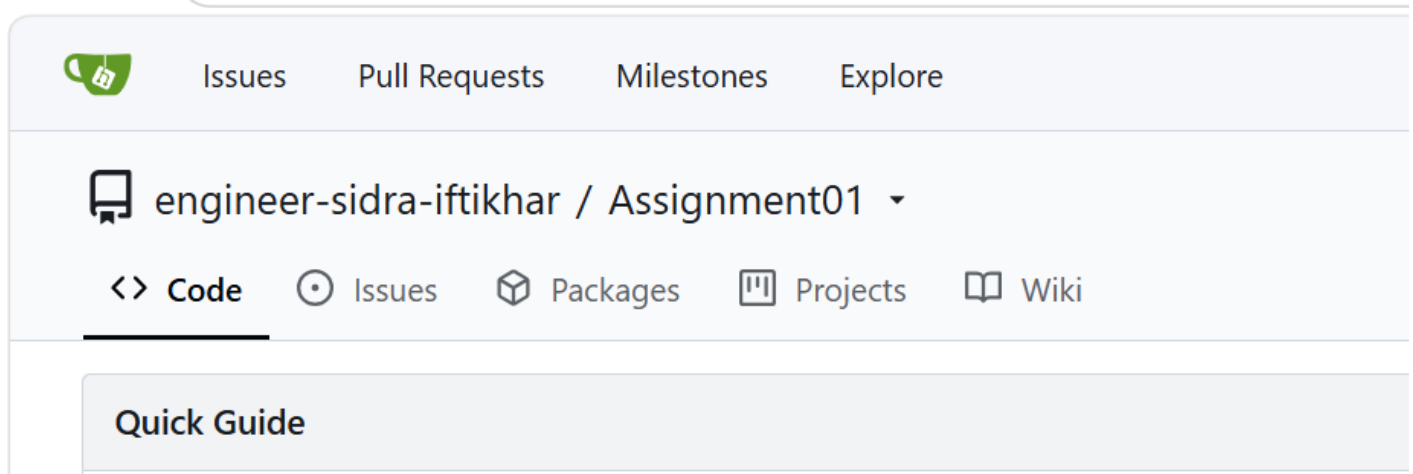


Fig. 11: Empty Assignment01 repository in Gitea

3. On the same Ubuntu server, create a separate local repository outside both the Gitea directory and the existing CC repository:

```
cd ~
mkdir Assignment01
cd Assignment01
git init
git branch -M main
```

```
engineer-sidra-iftikhar@ubuntu:~/Gitea$ cd ~
engineer-sidra-iftikhar@ubuntu:~$ mkdir Assignment01
engineer-sidra-iftikhar@ubuntu:~$ cd Assignment01
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/engineer-sidra-iftikhar/Assignment01/.git/
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git branch -M main
engineer-sidra-iftikhar@ubuntu:~/Assignment01$
```

Fig. 12: Local repository initialised on branch main

4. Create a README.md containing:

- your full name; and
- your registration number.

```

engineer-sidra-iftikhar@ubuntu:~/Assignment01$ printf "Full Name: Sidra Iftikhar\nRegistration Number: FA24B1-SE-063\n" > README.md
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ cat README.md
Full Name: Sidra Iftikhar
Registration Number: FA24B1-SE-063
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ |

```

Fig. 13: README.md created with name and registration number

5. Commit the README:

```

git add README.md
git commit -m "Add student information"

engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git config --global user.name "Sidra Iftikhar"
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git config --global user.email "engineersidra58@gmail.com"
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git add README.md
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) 2ee4135] Add student information
1 file changed, 2 insertions(+)
create mode 100644 README.md
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ |

```

Fig. 14: README committed to local repository

6. In Gitea, generate a personal access token:

- 🔗 Open your profile menu and select Settings.
- 🔗 Open Applications and then Manage Access Tokens.
- 🔗 Use ubuntu-server-token as the token name.
- 🔗 Give the token read-and-write repository permission (write:repository).
- 🔗 Copy the generated token. Do not place it in a command, remote URL, or screenshot.

Manage Access Tokens

These tokens grant access to your account using the Gitea API.

▼ Generate New Token

Token Name

ubuntu-server-token

Repository and Organization Access

☐ Public only

☒ All (public, private, and limited)

Fig. 15: Gitea access tokens page

repository	☐ No Access	☐ Read	☒ Read and Write
user	☒ No Access	☐ Read	☐ Read and Write

Generate Token

Fig. 16: Token permissions set to Read and Write

7. Because Gitea and Git are running on the same Ubuntu server, add the local Gitea address as the gitea remote. Include your Gitea username, but do not include the personal access token:

```
git remote add gitea http://@127.0.0.1:3000//Assignment01.git
```

The git remote add command only records the repository address; it does not contact Gitea or perform authentication.

8. Verify that the remote URL contains your username but does not contain a password or token:

```
git remote -v
```

```
engineer-sidra-iftikhar@ubuntu:~$ cd ~/Assignment01
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git remote add gitea http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git remote -v
gitea    http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (fetch)
gitea    http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (push)
engineer-sidra-iftikhar@ubuntu:~/Assignment01$
```

Fig. 17: Gitea remote added and verified

9. Push the initial commit:

```
git push -u gitea main
```

10. When Git displays the password prompt, paste the Gitea personal access token and press Enter: Password for 'http://@127.0.0.1:3000':

```
engineer-sidra-iftikhar@ubur x + v
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git remote -v
gitea    http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (fetch)
gitea    http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (push)
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://engineer-sidra-iftikhar@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 57.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
engineer-sidra-iftikhar@ubuntu:~/Assignment01$
```

Fig. 18: Initial commit pushed to Gitea

Step 1.3: Verify the Gitea Repository

1. Open the Assignment01 repository in Gitea.
2. Confirm that the repository owner and repository name are visible.
3. Confirm that the repository is public.
4. Confirm that the rendered README.md shows your full name and registration number. 5. Keep the Gitea URL visible in the browser.

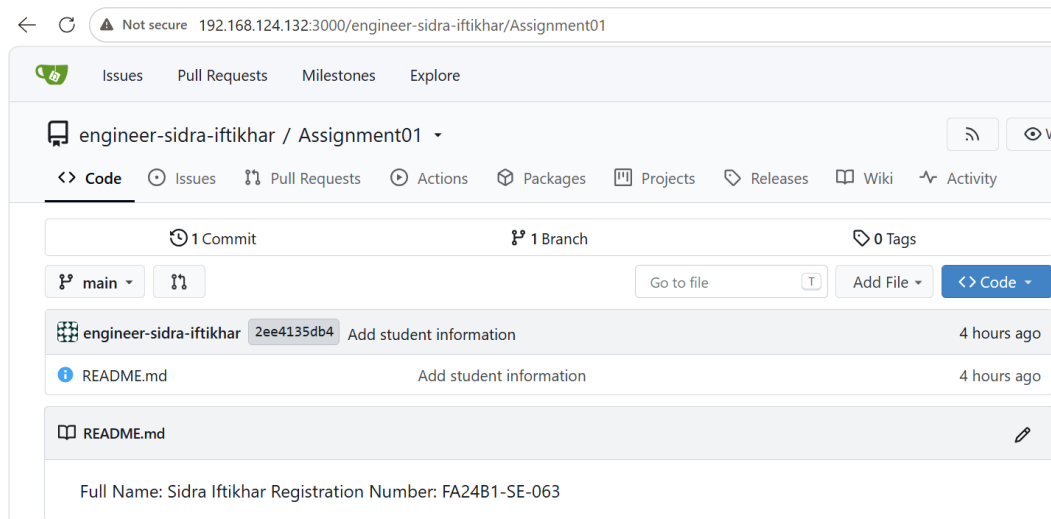


Fig. 19: Repository verified in Gitea (public)

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner * Repository name *

Assignment01 is available.

Great repository names are short and memorable. How about [silver-invention](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Add README ☐

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Add license

Licenses explain how others can use your code. [About licenses](#)

Create repository

Fig. 20: Creating the Assignment01 repository on GitHub

```

engineer-sidra-iftikhar@ubuntu:~/Assignment01$ cd ~/Assignment01
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git remote add github https://github.com/engineer-sidra-iftikhar/Assignment01.git
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git remote -v
gitea  http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (fetch)
gitea  http://engineer-sidra-iftikhar@127.0.0.1:3000/engineer-sidra-iftikhar/Assignment01.git (push)
github https://github.com/engineer-sidra-iftikhar/Assignment01.git (fetch)
github https://github.com/engineer-sidra-iftikhar/Assignment01.git (push)
engineer-sidra-iftikhar@ubuntu:~/Assignment01$

```

Fig. 21: Both Gitea and GitHub remotes verified

Step 2.2: Push to GitHub

```

engineer-sidra-iftikhar@ubur  ×  +  ▾
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': engineer-sidra-iftikhar
Password for 'https://engineer-sidra-iftikhar@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 96.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/engineer-sidra-iftikhar/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
engineer-sidra-iftikhar@ubuntu:~/Assignment01$

```

Fig. 22: Pushing the repository to GitHub

Step 2.3: Verify the GitHub Repository

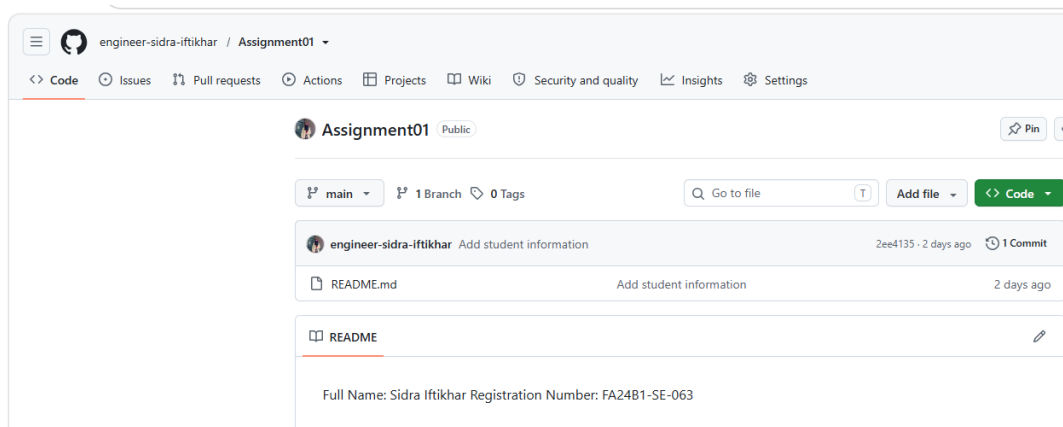


Fig. 23: Repository verified on GitHub

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```

engineer-sidra-iftikhar@ubur x + v
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ |

```

Fig. 24: Git LFS installed and *.bin tracked

Step 3.2: Create and Verify Three LFS Files

```

engineer-sidra-iftikhar@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ |

```

Fig. 25: Three large files tracked with Git LFS

Step 3.3: Commit and Push the LFS Files

```

engineer-sidra-iftikhar@ubuntu:~/Assignment01$ git push github main
^[[DUpsloading LFS objects: 0% (0/3), 296 MB | 579 KB/s
^[[CUpsloading LFS objects: 0% (0/3), 302 MB | 582 KB/s
Uploading LFS objects: 100% (3/3), 321 MB | 574 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 799 bytes | 266.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/engineer-sidra-iftikhar/Assignment01.git
2ee4135..431ce1b main -> main
engineer-sidra-iftikhar@ubuntu:~/Assignment01$ |

```

Fig. 26: LFS files pushed to GitHub

Task 4: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository

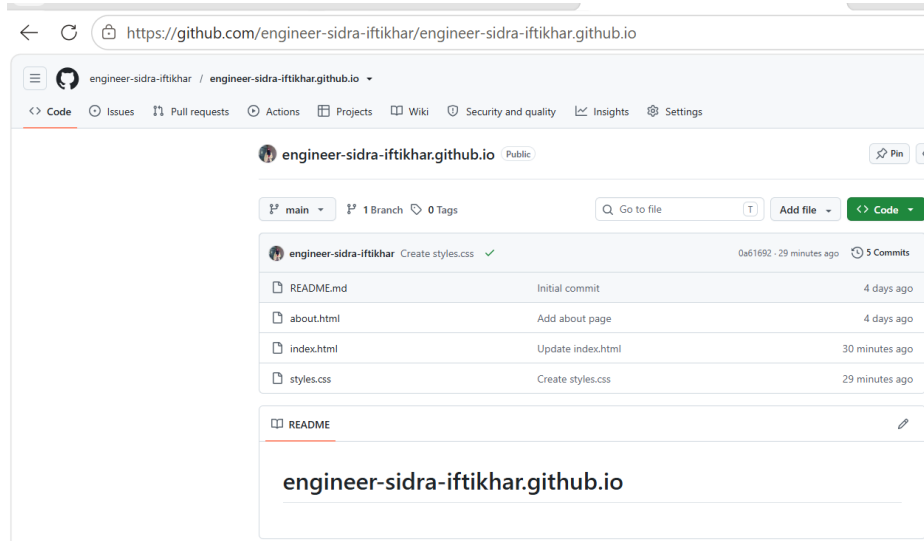


Fig. 27: Portfolio repository on GitHub

GitHub Pages

GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.

Your site is live at <https://engineer-sidra-iftikhar.github.io/>
Last deployed by [engineer-sidra-iftikhar](#) 30 minutes ago

Visit site

Unpublish site

Build and deployment

Source

Deploy from a branch

Branch

Your GitHub Pages site is currently being built from the `main` branch. [Learn more about configuring the publishing source for your site.](#)

main

/ (root)

Save

Learn how to [add a Jekyll theme](#) to your site.

Your site was last deployed to the `github-pages` environment by the `pages build and deployment` workflow.

[Learn more about deploying to GitHub Pages using custom workflows](#)

Fig. 28: GitHub Pages site published

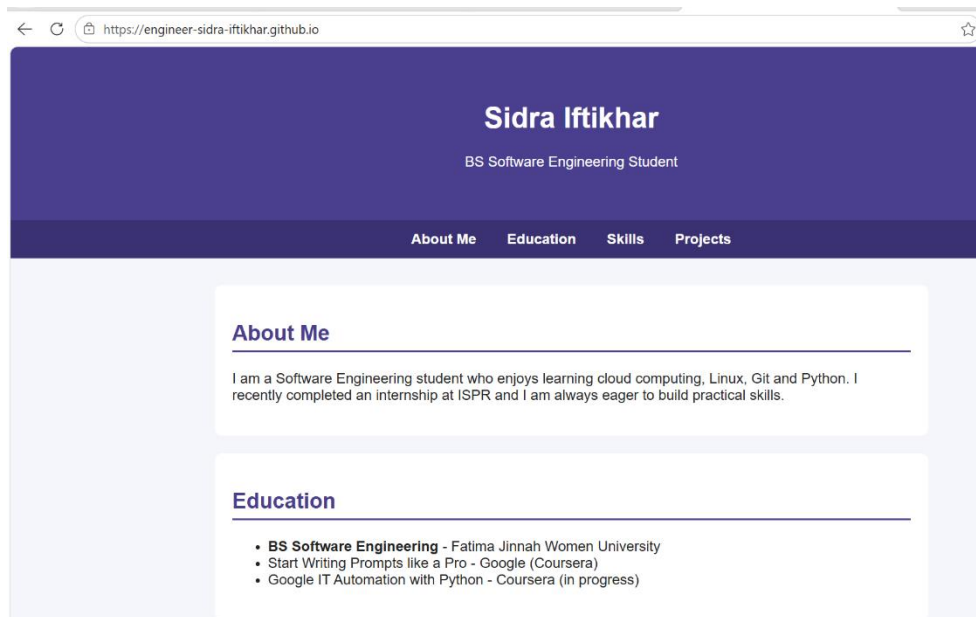


Fig. 29: Portfolio home page header

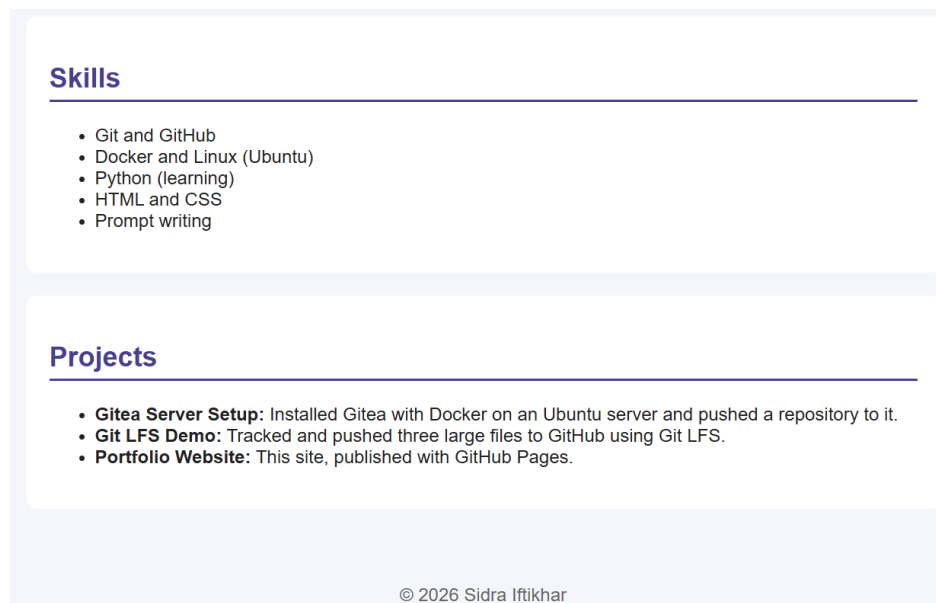


Fig. 30: Portfolio Skills section